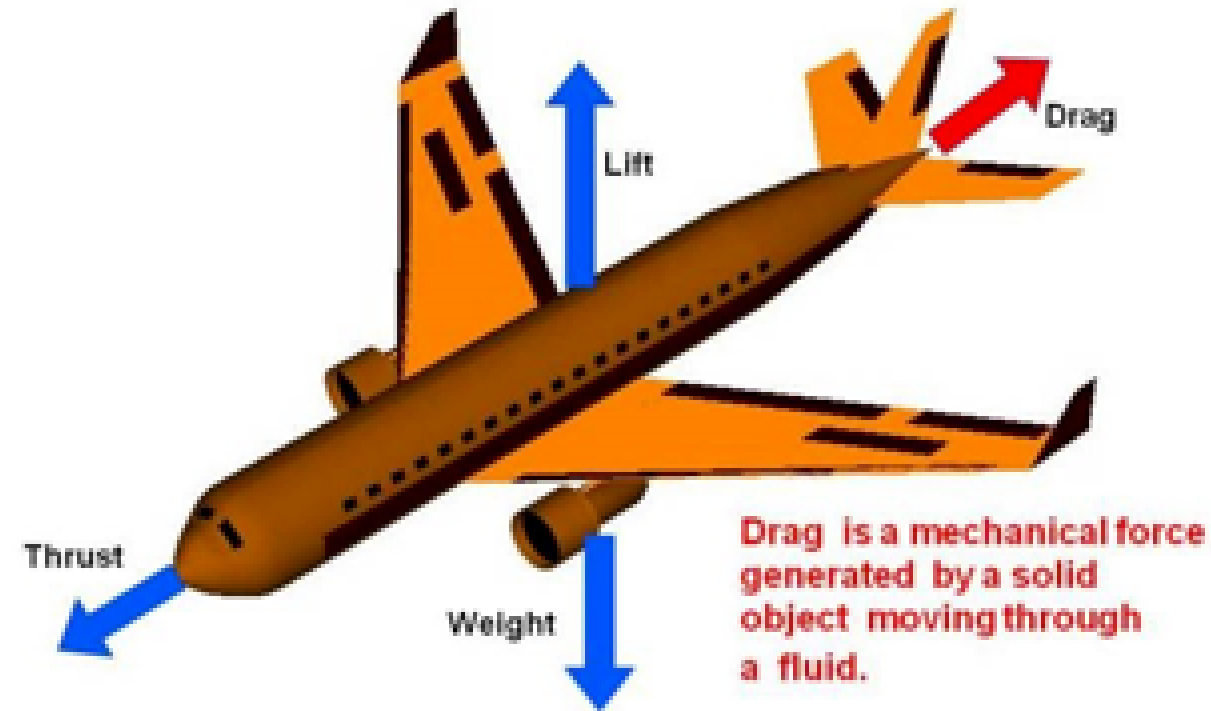


Day 07 - Drag Forces



Big Announcement

MSU Tenure System Faculty Union Recognized

What does that mean for
you?

- Faculty can bargain with the university for what's needed to meet our educational needs
- For example, there's more of y'all every year, but not more of us.

WE MADE HISTORY!



**WE ARE
RECOGNIZED**

Announcements

- Homework 2 is due Friday; late after Sunday 11:59pm (always)
- Homework 3 is now posted; due next Friday; late after Sunday...
- Office hours, for now (Mihir-MN; Danny-DC):
 - Tuesday 6-8pm (MN, Zoom)
 - Thursday 6-8pm (MN, Zoom)
 - Friday 2-4pm (DC, 1248 BPS)
- Zoom Link: <https://msu.zoom.us/j/96882248075>
 - password: phy321msu
- Friday's Class: **Will work Exercises 3a, 5a, & 5b form HW 3 together + questions**

Seminars this week

WEDNESDAY, September 10, 2025

Astronomy Seminar, 1:30 pm, 1400 BPS, In Person and Zoom, Host~

Speaker: Rachael Roettenbacher, University of Michigan

Title: Imaging Spotted Stars for an Improved Understanding of Stars and Exoplanets

Zoom Link: [https://msu.zoom.us/j/887295421?](https://msu.zoom.us/j/887295421?pwd=N1NFb0tVU29JL2FFSkk0cStpanR3UT09)

[pwd=N1NFb0tVU29JL2FFSkk0cStpanR3UT09](https://msu.zoom.us/j/887295421?pwd=N1NFb0tVU29JL2FFSkk0cStpanR3UT09)

Meeting ID: 887-295-421

Passcode: 002454

Seminars this week

WEDNESDAY, September 10, 2025

FRIB Nuclear Science Seminar, 3:30pm., FRIB 1300 Auditorium and online via Zoom

Speaker: Suzanne Lapi of the University of Alabama at Birmingham

Title: Development of new isotopes for theranostic applications

Please see website for full abstract.

Please click the link below to join the webinar:

Join Zoom: [https://msu.zoom.us/j/96485010083?
pwd=O0rXwspn80aYGEl06QEZag6Ao4siq7.1](https://msu.zoom.us/j/96485010083?pwd=O0rXwspn80aYGEl06QEZag6Ao4siq7.1)

Meeting ID: 964 8501 0083

Passcode: 261744

Seminars this week

THURSDAY, September 11, 2025

Colloquium, 3:30 pm, 1415 BPS, in person and zoom. Host ~

Refreshments and social half-hour in BPS 1400 starting at 3 pm

Speaker: Laura Chomiuk, MSU

Title: Fall 2025 Physics and Astronomy Kickoff

Background:

For more information and to schedule time with the speaker, see the colloquium calendar at <https://pa.msu.edu/news-events-seminars/colloquium-schedule.aspx>

Zoom Link: <https://msu.zoom.us/j/94951062663>

Password: 2002 Or complete link: [https://msu.zoom.us/j/94951062663?](https://msu.zoom.us/j/94951062663?pwd=c48uM25P9UsRVuR74rkOioOWgpoxgC.1)

[pwd=c48uM25P9UsRVuR74rkOioOWgpoxgC.1](https://msu.zoom.us/j/94951062663?pwd=c48uM25P9UsRVuR74rkOioOWgpoxgC.1)

Seminars this week

FRIDAY, September 12, 2025

QuIC Seminar, 12:30pm, -1:30pm, 1300 BPS, In Person

Speaker: Jean Paul Sadia, MSU

Title: Introduction to Quantum Information and Computation

Full Scheule is at: <https://sites.google.com/msu.edu/quic-seminar/>

For more information, reach out to Ryan LaRose

Seminars this week

FRIDAY, September 12, 2025

IReNA Online Seminar, 2:00 pm, via Zoom.

Hosted by: Artemis Tsantiri (University of Regina, Canada)

Speaker: Lorenzo Roberti, INFN-LNS, Italy/Konkoly, Observatory Hungary

Title: Carbon-Oxygen Shell Mergers in Massive Stars

Zoom Link: <https://msu.zoom.us/j/827950260>

Password: JINA

AI Policy Proposals

- **Proposal 1:** We adopt a policy that does not allow AI use at all.
- **Proposal 2:** We adopt a policy that allows AI use for brainstorming, help, and editing.
- **Proposal 3:** We adopt a policy that allows AI for use in nearly any way.
- **Proposal 4:** We adopt a policy that allows AI for use in any way with no documentation required.

Updated AI Policy

20 Responses



Proposal 2

We adopt a policy that allows AI use for brainstorming, help, and editing.

- AI cannot be used for direct answers or completion of assignments.
- We expect documentation of AI use, but it can be informal.
- Violations are discussed with Danny; the first violation requires a redo of the assignment, and repeated violations result in a failing grade.

Received 75% of first choice votes

Goals for Week 3

- Be able to answer the following questions.
 - What is Mathematical Modeling?
 - What is the process for analyzing these models?
- Be able to solve "Simple" Motion Problems with Newton's Laws.

Our man, Reynolds

- The Reynolds number is a dimensionless quantity.
- It is a ratio of inertial forces to viscous forces.

$$Re = \frac{\rho v L}{\mu}$$

- ρ - density of the fluid
- v - velocity of the object
- L - characteristic length
- μ is the dynamic viscosity



Our man, Reynolds

BTW, this is not a photo of Reynolds.

- This is Stokes.
 - He developed the concept of the Reynolds number.
 - Reynolds "popularized" it according to the Wikipedia.

$$Re = \frac{\rho v L}{\mu}$$

Discussion: What kinds of systems have a high/low Reynolds number?



Clicker Question 6-2a

I made a mistake explaining our linear drag model derivation.

Consider a ball falling down with a coordinate system such that y is positive downward (as before). Which of the following is correct for the differential equation:

1. $m\ddot{y} = -mg + bv_y$

2. $m\ddot{y} = -mg - bv_y$

3. $m\ddot{y} = +mg + bv_y$

4. $m\ddot{y} = +mg - bv_y$

Think about the SIGN of v_y ; it's a component not a speed!

Clicker Question 6-2b

Assuming a **linear model** for Air Resistance $\sim bv$, we can obtain this EOM for the velocity a falling ball, after *separation of variables*:

$$\frac{dv_y}{v_y - v_{term}} = -\frac{b}{m}dt$$

Assume we try to integrate this equation from $v_y = v_{y0}$ at $t = 0$ to $v_y(t)$ at time t .

Write the integral (on both sides) that is needed to solve this equation.

This integral setup is common in physics derivations with ODEs; provided we can find an anti-derivative.

Clicker Question 6-3

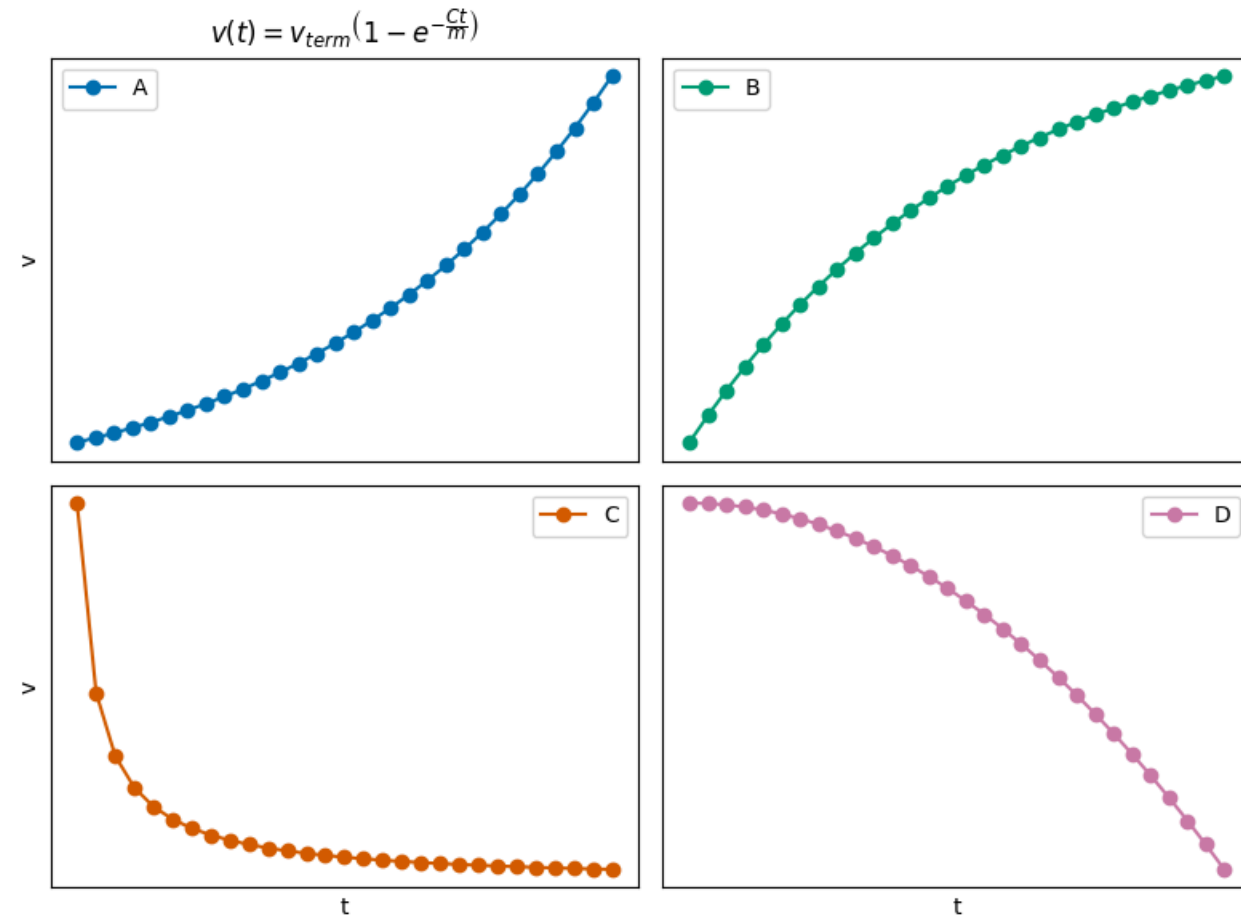
For the system of **Linear Drag in 1D**, we found a solution for the velocity as a function of time, with $v = 0$ at $t = 0$.

$$v(t) = v_{term} \left(1 - e^{-\frac{bt}{m}} \right)$$

where $v_{term} = \sqrt{\frac{mg}{b}}$.

CQ 6-3

Which sketch could be correct for the velocity of the ball?



Clicker Question 6-4

For the system of **Quadratic Drag in 1D**, we found a solution for the velocity as a function of time, with $v = 0$ at $t = 0$.

$$v(t) = v_{term} \tanh(gt/v_{term})$$

where $v_{term} = (mg/c)^{1/2}$. Do the units make sense? What are the units $[gt/v_{term}]$?

1. Yes, v and $(mg/c)^{1/2}$ **have the same units**; the units for $[gt/v_{term}]$ are m/s.
2. No, v and $(mg/c)^{1/2}$ **have different units**; the units for $[gt/v_{term}]$ are m/s.
3. Yes, v and $(mg/c)^{1/2}$ **have the same units**; the units for $[gt/v_{term}]$ are unit-less.
4. No, v and $(mg/c)^{1/2}$ **have different units**; the units for $[gt/v_{term}]$ are unit-less.

Clicker Question 6-5

For the system of **Quadratic Drag in 1D**, we found a solution for the velocity as a function of time, with $v = 0$ at $t = 0$.

$$v(t) = v_{term} \tanh(gt/v_{term})$$

where $v_{term} = \sqrt{mg/c}$. What happens when $t \rightarrow \infty$?

1. The object stops moving.
2. The object travels at a constant velocity.
3. The object travels at an increasing velocity.
4. The object travels at a decreasing velocity.
5. I'm not sure.